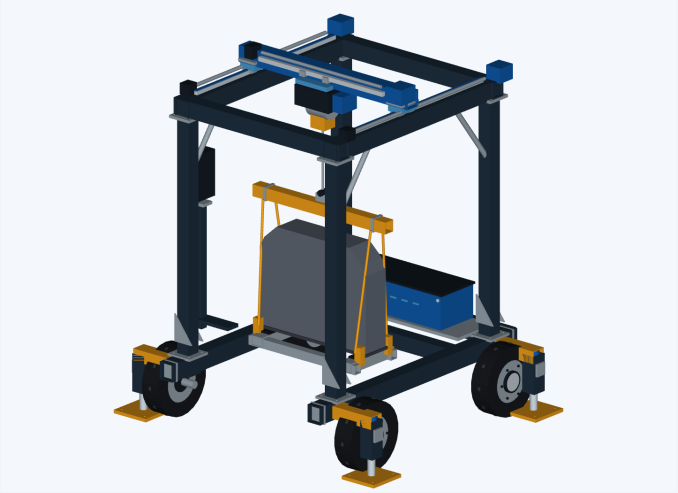
Two different stones, two purposes

Updated exports from the existing RoverSetter design models; two stone representations show different project contexts.

The complete original-design CAD model shows the monument-handling architecture. The Prototype 3 physical test used a smaller stone, recalled as 16 x 8 x 6 inches and approximately 70 lb, supported in front of the wheel line.

A | COMPLETE DESIGN MODEL



Complete design CAD with 845 x 190 x 680 mm monument geometry and positioning components. Not a photograph of the prototype.

B | PROTOTYPE 3 CAD VIEW



16 x 8 x 6-inch test-stone geometry shown with the original design frame; hoist and carriage are hidden in this CAD view.

STONE DIMENSIONS

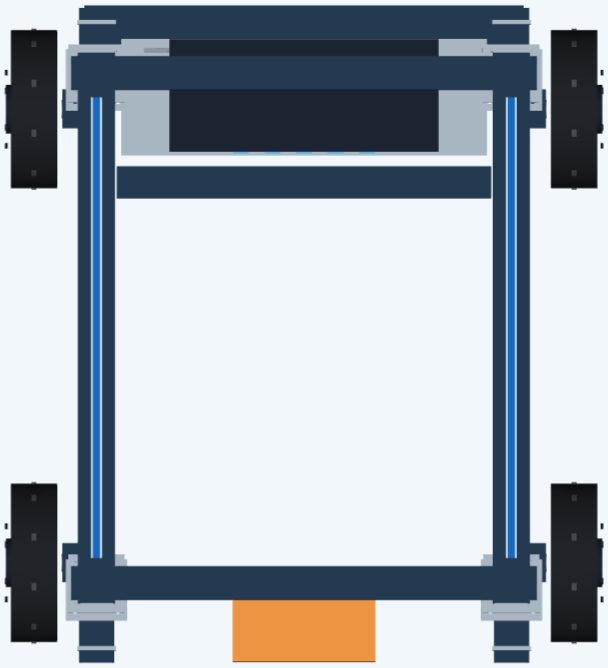
Dimension	Complete design view	Prototype 3 test stone
Across machine	845 mm	406.4 mm (16 in)
Front to back	190 mm	203.2 mm (8 in)
Vertical	680 mm	152.4 mm (6 in)
Weight	Not verified (~600+ lb if granite)	~70 lb recalled

WHAT THE VIEWS REPRESENT

These are updated exports of the designer’s existing RoverSetter models. The simplified view hides suspension and positioning components to show the frame and test-stone size. The original test’s precise hook location and stone coordinates were not measured; no load rating is claimed.

Prototype 3 frame and test stone

Top-view CAD export and design dimensions; exact physical test pose is not recorded.



DIMENSIONS SUPPORTED BY OWNER RECOLLECTION

Stone: 406.4 mm transverse (left-right) x 203.2 mm front-back x 152.4 mm tall. Nominal rectangular volume: 768 in³ (0.012585 m³). At a recalled 70 lb, implied density is approximately 2.52 g/cm³, physically plausible for natural stone.

ORIGINAL DESIGN DIMENSIONS IN CURRENT CAD

Chassis 1,360 x 1,950 mm; wheel centers 1,536 mm transverse and 1,290 mm fore-aft; wheel diameter 450 mm; gantry top 2,050 mm. The designer confirms that these are original design dimensions preserved in the current exports.

The CAD view documents design geometry, not the recorded location of every component during the historic trial. The physical test overhang and hook position were not measured; the current stone coordinates do not establish either value.

What the drawing does—and does not—prove

Two distinct observed stages; gravity-only static analysis does not explain powered-startup rocking.

OBSERVED SEQUENCE | USER RECOLLECTION

1. STATIONARY

Approx. 70-lb load suspended forward of the front wheels on a hard surface; wheel-only support.

2. REAR UNLOADING

Both rear wheels lifted nearly together (one slightly higher). Machine rocked and settled back.

3. MOTORIZED STARTUP

Pitching increased immediately when drive motors started and the chassis accelerated forward.

WHAT THE EQUATIONS ACTUALLY ADDRESS

$$N_{rear}(d) = [\, M\, g\, L - m\, g\, d \,] / B$$

For a fixed rigid machine on level ground, increasing forward load overhang d decreases predicted static rear reaction. Model masses, restoring arm and original overhang have not been measured; no historical tipping threshold is established.

WHAT REMAINS OPEN

A powered forward acceleration does not by itself prove a greater forward gravitational tipping moment. The additional motor-startup pitching may involve suspended-load motion, varying contacts, structural compliance, motor reaction, or a transient rocking response. None has been established as the cause.

STATUS OF THE CAD ARTIFACT

The STEP is an updated CAD export of the designer’s RoverSetter model. The simplified display does not show every mechanism, nor does it establish physical test pose or lifting capacity. Exact machine mass properties, actual test overhang and dynamic response remain unmeasured.